

Symmetry and skew



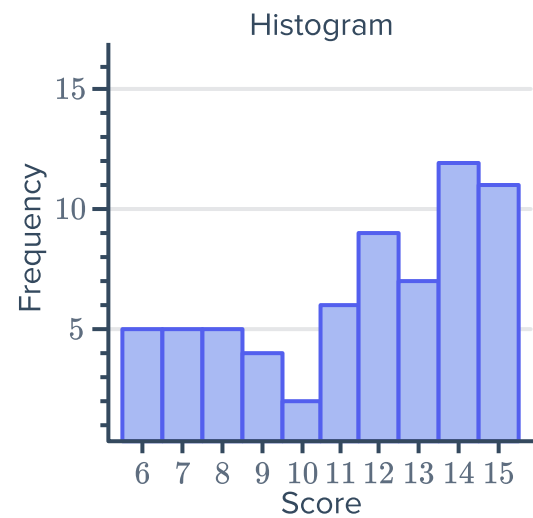
1 Describe the shape of the data in the following graphs:

a

Leaf	
1	6 7 7
2	2 2 2 2 3 3 3
3	3 3 3 6 6 6 7 7 7 7 7
4	4 4 4 4 4 4
5	7 7

Key: 2|3 = 23

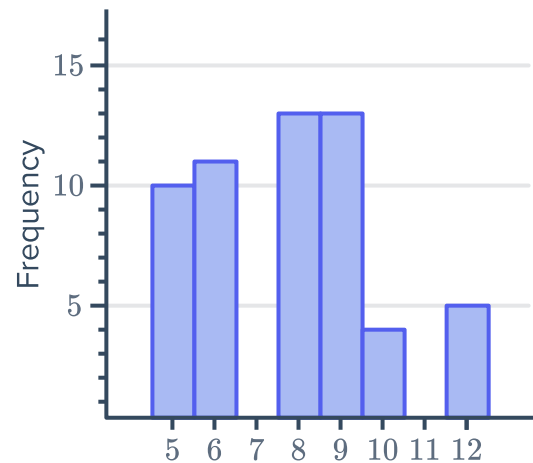
b



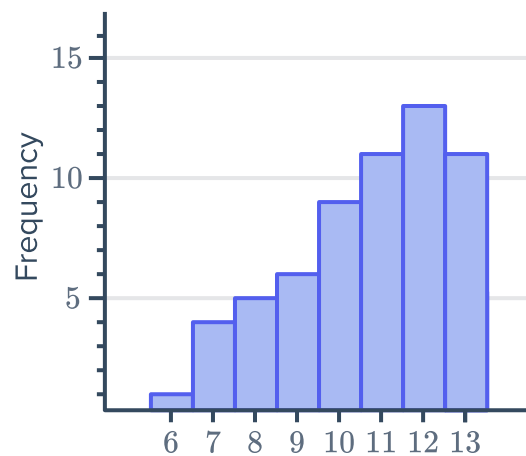
c



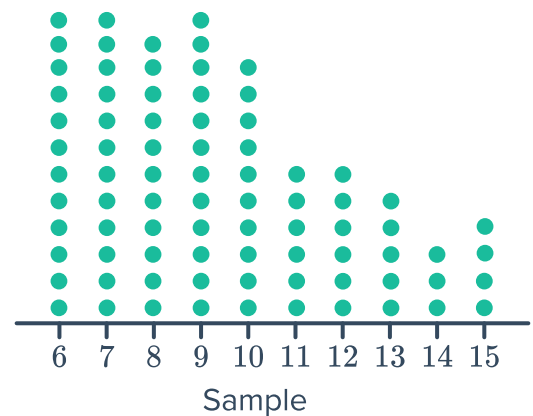
d



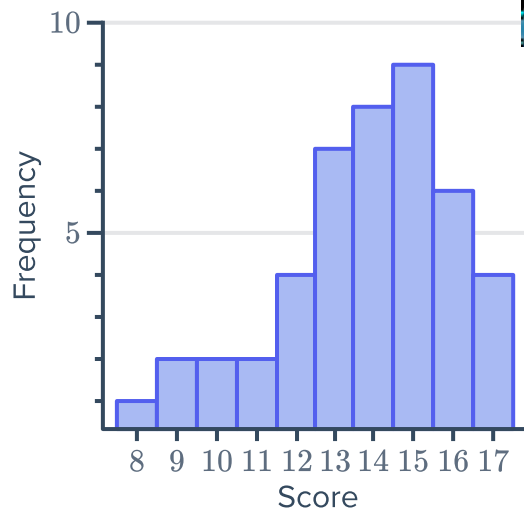
e



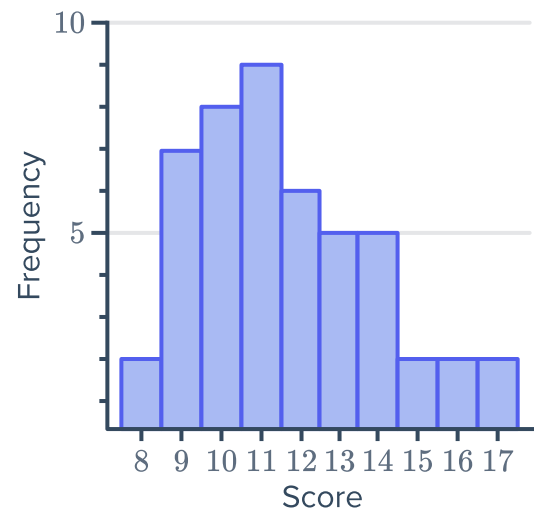
f



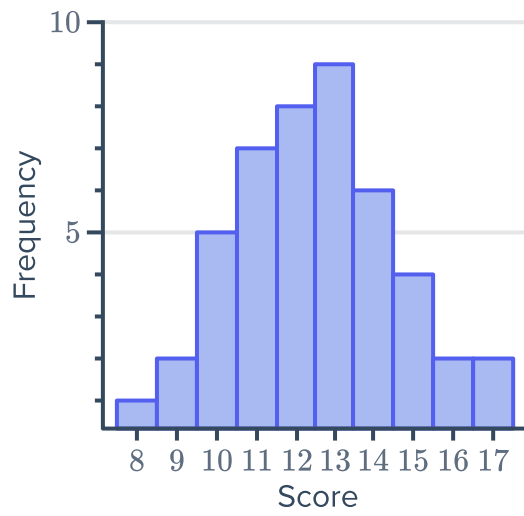
g



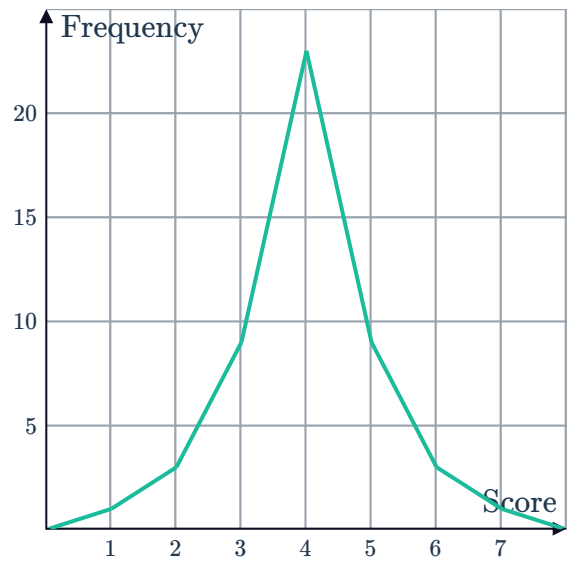
h



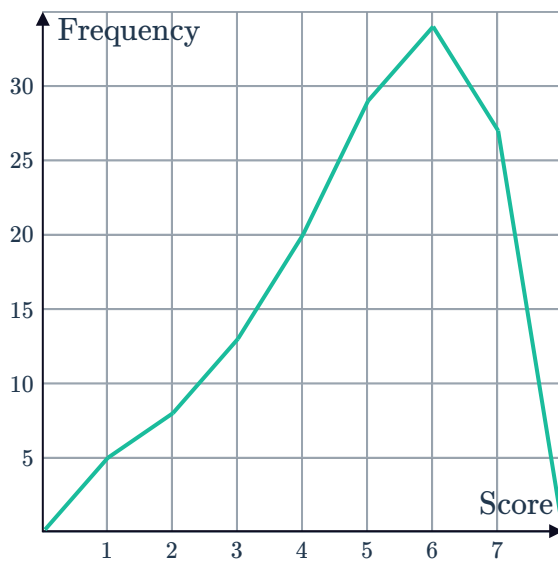
i



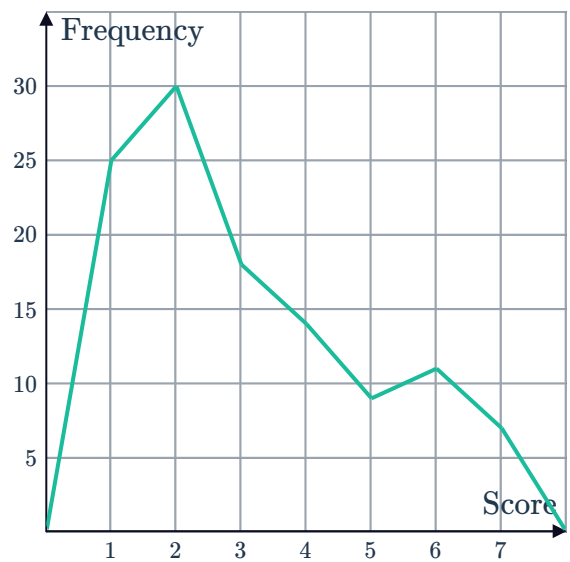
j



k



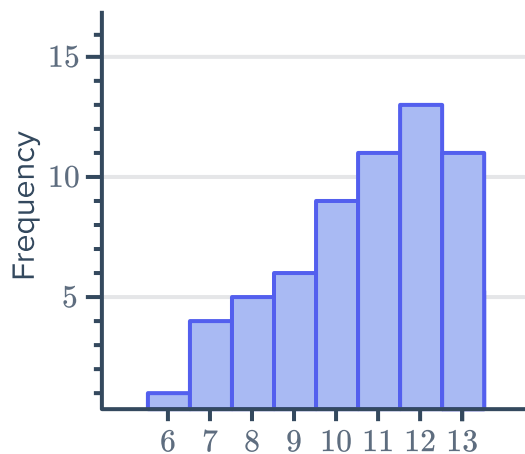
l



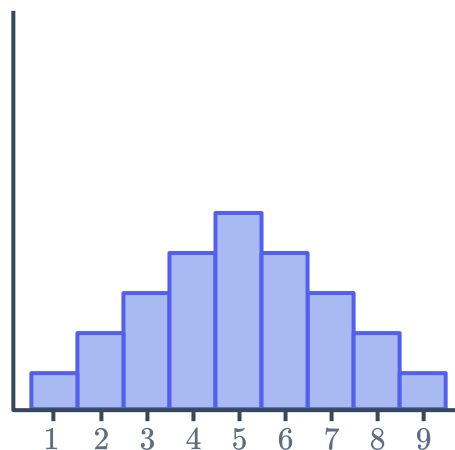
2 Determine whether the following graphs are modal:



a



b



c

Leaf	
1	0 0 2 7
2	2 2 3 3 5 8
3	1 6 4
4	7
5	0 1 6
6	5 7 7 8 8
7	4 4 4
8	4 4

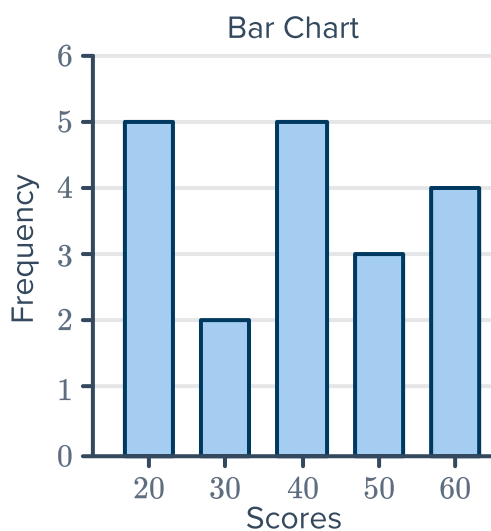
Key: 2|3 = 23

d

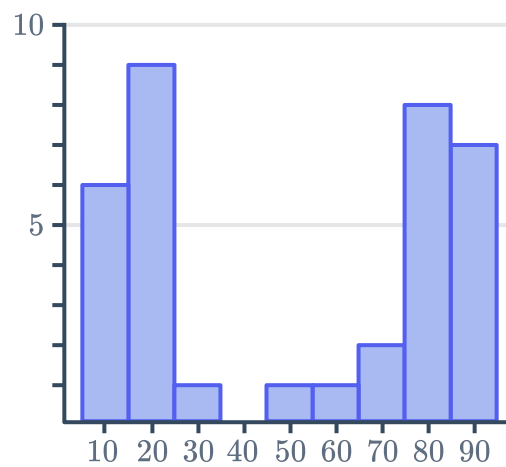
Leaf	
1	2 3 4 5 5 6 6 7 7 9
2	1 2 4 4
3	0 2 9 9
4	
5	5

Key: 1|2 = 12

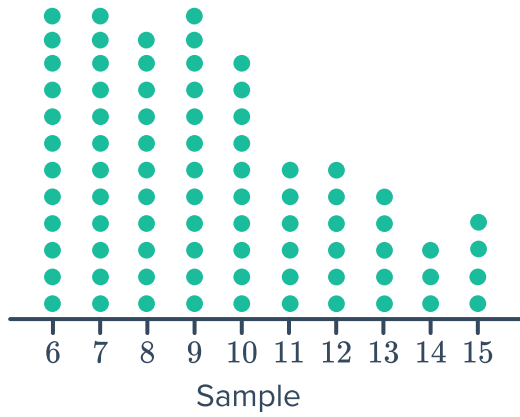
e



f

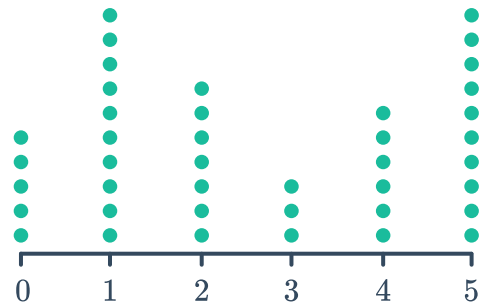


g

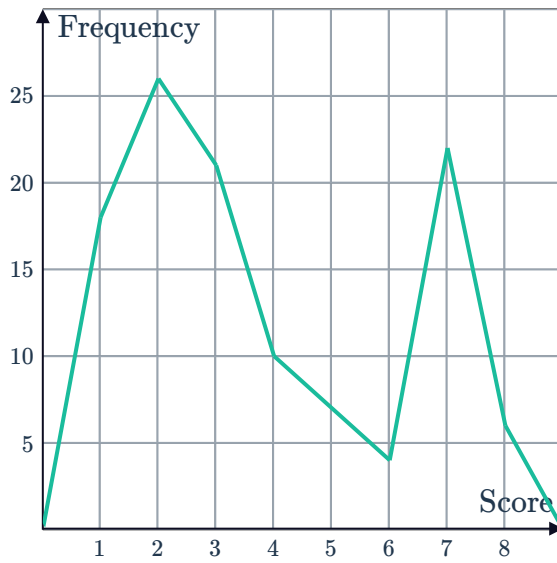


h

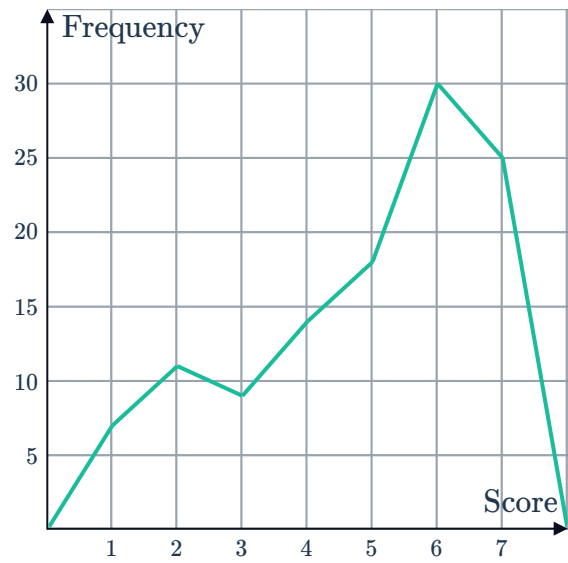
Number of goals scored



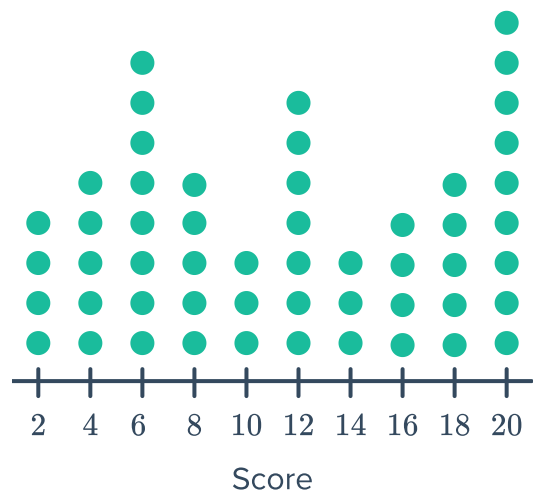
i



j



3 How would you describe the modality of the following dot plot? Explain your answer.



- 4 The table shows the number of crime novels in a bookshop for different price ranges rounded off to the nearest \$5:



Price of crime novel	Frequency
\$5	5
\$10	10
\$15	17
\$20	8
\$25	17
\$30	10
\$35	5

- a Graph this data as a histogram.
- b Describe the shape of the distribution of the data.

Clusters and outliers

- 5 The number of hours worked per week by a group of people is represented in the following stem-and-leaf plot:

- a State the value of any outliers.
- b Is there any clustering of data? If so, in what interval?
- c State the mode.

	Leaf
0	2
1	
2	0 3 6 6
3	1 4 5 6 6 7
4	0 4 6 7 9
5	0

Key: $2|3 = 23$

- 6 Consider the stem plot given:

- a State the value of any outliers.
- b Is there any clustering of data? If so, in what interval?
- c State the mode.
- d Describe the shape of the distribution.

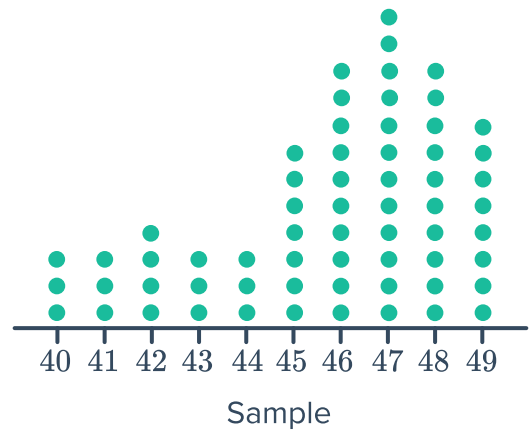
	Leaf
0	5
1	7 8
2	0 8
3	1 3 3 7 8 9
4	1 3 5 8 8 8
5	
6	
7	
8	
9	2

Key: $2|3 = 23$

- 7 Temperatures were recorded over a period of time and presented as a dot plot:



- Are there any outliers?
- Is there any clustering of data? If so, in what interval?
- What is the modal temperature?
- Describe the shape of the distribution.



- 8 The number of peanuts in mixed nut packets were sampled and recorded in the following stem plot:

- Complete the frequency distribution table:

Score	Frequency
40 – 49	
50 – 59	
60 – 69	
70 – 79	
80 – 89	
90 – 99	
100 – 109	
110 – 119	

	Leaf
4	3 6 8
5	1 2 2 6 7 7 8
6	0 0 2
7	3 3 4 5 9
8	1 1 1 4 6 8 8 9
9	0 2 5 6
10	1 2 3 5 5 6 7 8
11	0 4 5 7

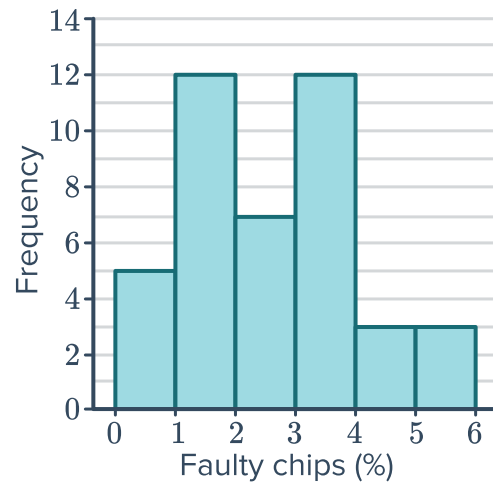
Key: 5|2 = 52

- Describe the modality of the distribution.
- State the modal class or classes of the data.

- 9 The percentage of faulty computer chips in 42 batches were recorded in the given histogram:



- Describe the modality of the distribution.
- State the modal class or classes of the data.



- 10 Consider the dot plot below:



- Are there any outliers?
- Is there any clustering of data?
- State the modal score(s).
- Describe the shape of the distribution.

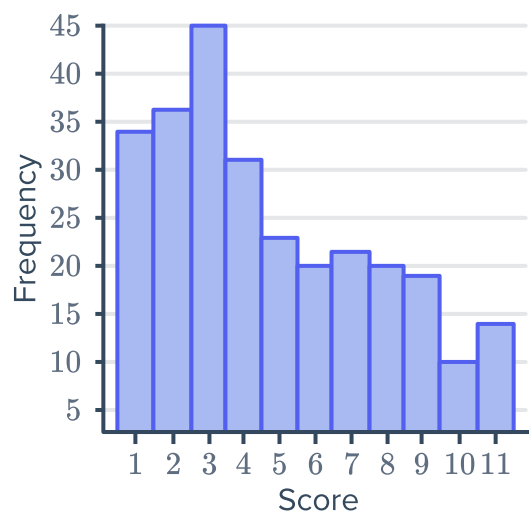
11 The reaction time of drivers was tested and recorded in the dot plot below:



- a Construct a frequency distribution table for the individual data values.
- b Describe the modality of the distribution.
- c State the mode(s).

12 Consider the data shown in the histogram:

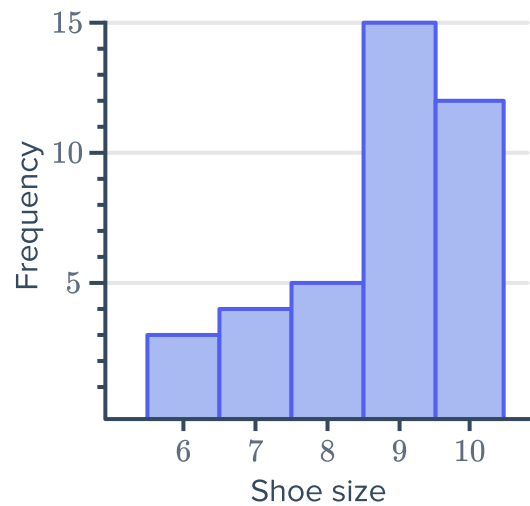
- a Are there any outliers?
- b Is there any clustering of data? If so, in what interval?
- c State the mode.
- d Describe the shape of the distribution.



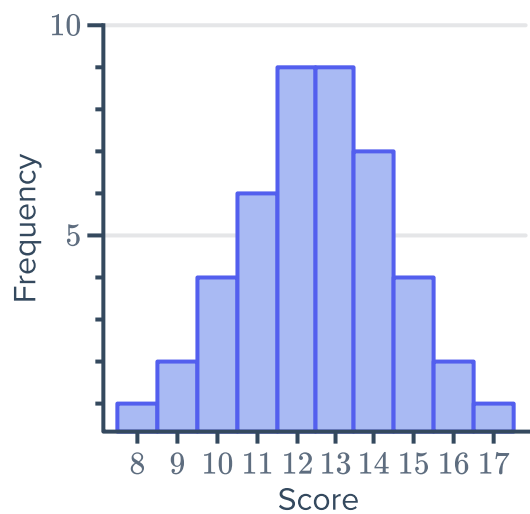
13 The shoe sizes of all the students in a class were measured and the data was presented in a graph:



- a** Are there any outliers?
- b** Is there any clustering of data? If so, in what interval?
- c** What is the modal shoe size?
- d** Describe the shape of the distribution.

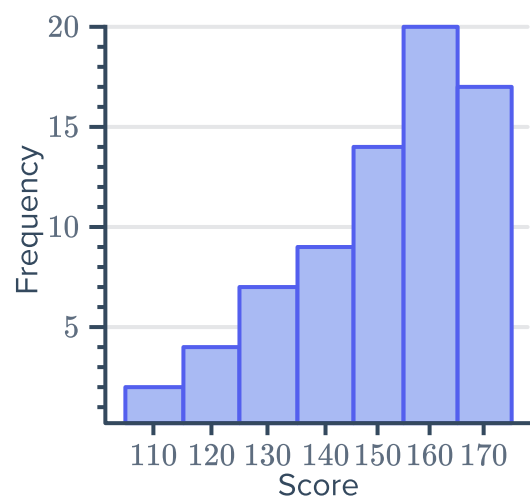


14 Estimate the value of the mean of the following data set correct to one decimal place:



15 Consider the histogram representing students' heights in centimetres:

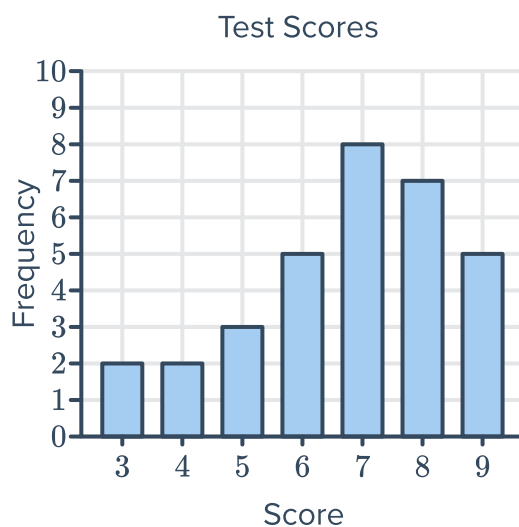
- a** Does the histogram most likely represent grouped data or individual scores?
- b** Estimate the value of the mean to one decimal place.
- c** Describe the shape of the distribution.



16 Consider the given column graph:



- a Describe the shape of the distribution.
- b Find the following:
 - i Lower quartile
 - ii Upper quartile
- c Hence, calculate the interquartile range.
- d Are there any outliers? If so, state the value.



17 Consider the dot plot given:

- a Describe the shape of the distribution.
- b Find the following:
 - i Lower quartile
 - ii Upper quartile
- c Hence, calculate the interquartile range.
- d Are there any outliers? If so, state the value.



18 The stem-and-leaf plot below shows the age of people to enter through the gates of a concert in the first 5 seconds:

- a Find the median age.
- b Find the difference between the lowest age and the median.
- c Find the difference between the highest age and the median.
- d Calculate the mean age, correct to two decimal places.
- e Is the data positively or negatively skewed?

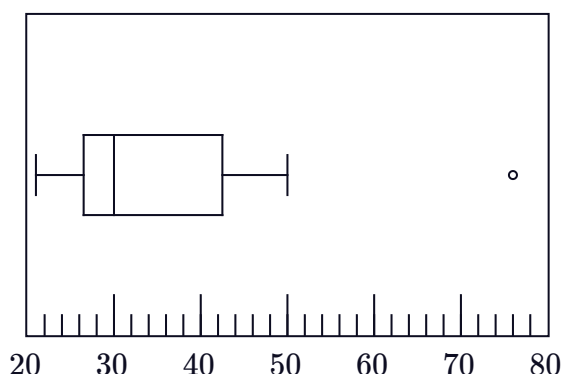
Leaf	
1	0 1 2 2 3 3 4 4 4 8 8 8
2	1 7
3	4 5 5
4	0
5	4

Key: 1|2 = 12 years old

- 19  $VO_2\text{Max}$ is a measure of how efficiently your body uses oxygen during exercise. The more physically fit you are, the higher your $VO_2\text{Max}$. Here are some people's results, listed in ascending order, when their $VO_2\text{Max}$ was measured:

21, 21, 23, 25, 26, 27, 28, 29, 29, 29, 30, 30, 32, 38, 38, 42, 43, 44, 48, 50, 76

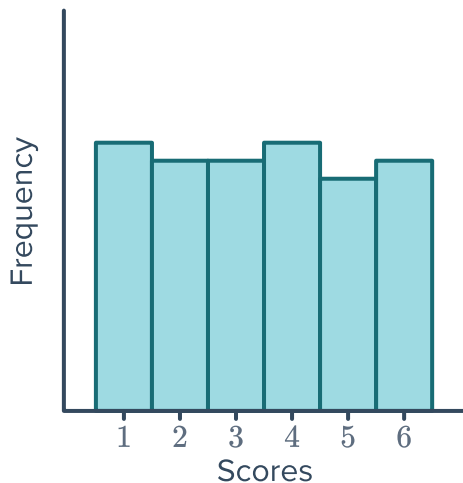
- Determine the median $VO_2\text{Max}$.
- Determine the upper quartile value.
- Determine the lower quartile value.
- Consider the box plot for this data set and state whether the results are positively or negatively skewed.
- State the value of the outlier.
- An average untrained healthy person has a $VO_2\text{Max}$ between 30 and 40. What can we say about the exercise habits of the majority of this group of people?



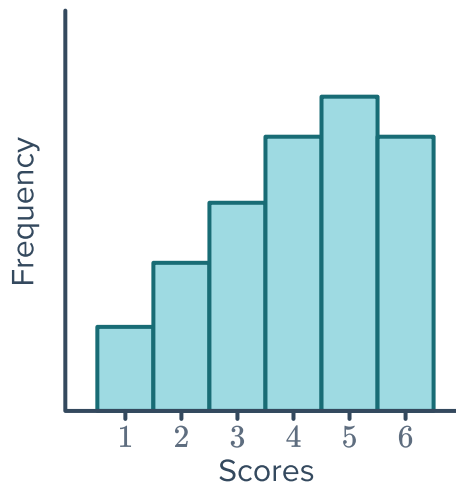
- 20 A die is rolled for a large number of trials and the number appearing is noted.

- Which histogram would you expect to match the data? Explain your answer.

A



B



- Describe the shape of the distribution of the data chosen above.

21 A pair of dice are rolled and the numbers appearing on the uppermost face are added to create a score.



a How many combinations would result in a score of:

i 4

ii 7

iii 10

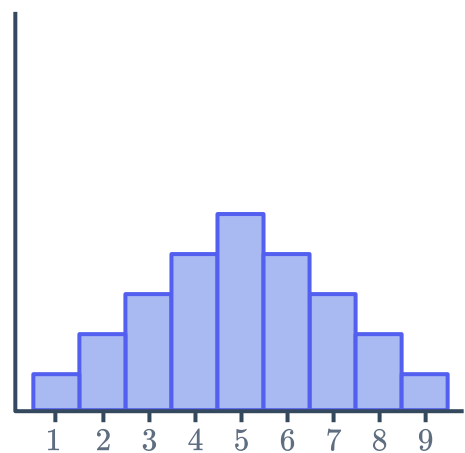
b A pair of dice are rolled for a large number of trials and the numbers appearing are added to create a score. Draw a sketch of a histogram you would expect to match the data.

Connect histograms and box plots

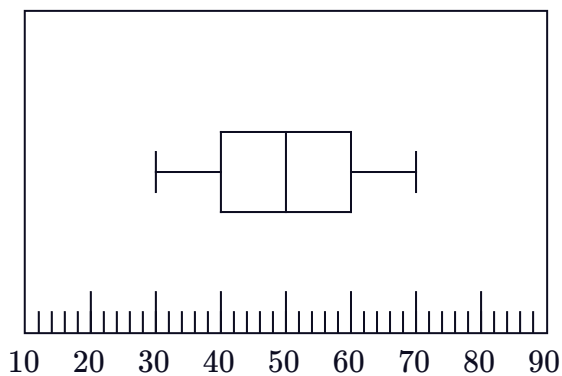
22 Match the histograms on the left to the corresponding box plots on the right:



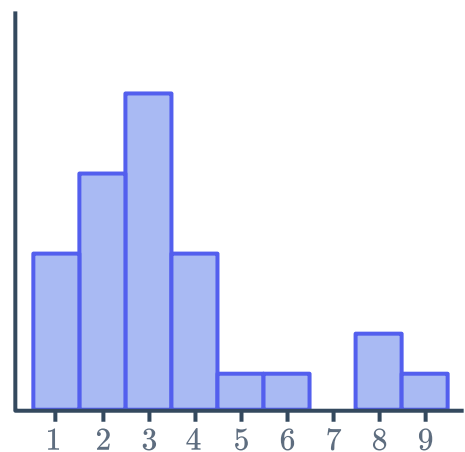
Histogram A



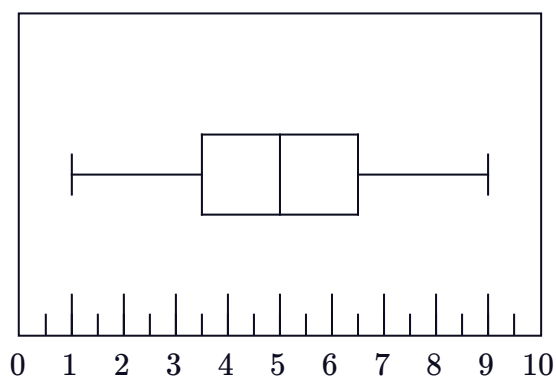
Box Plot 1



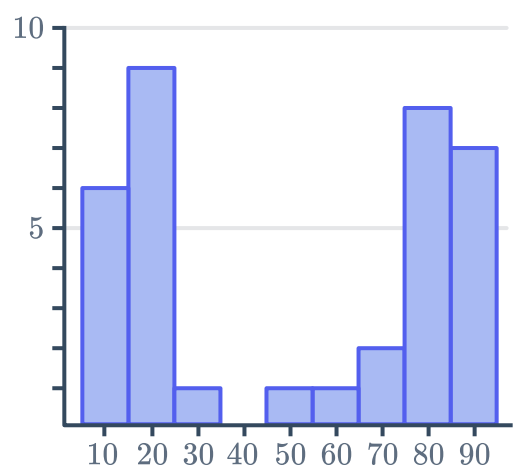
Histogram B



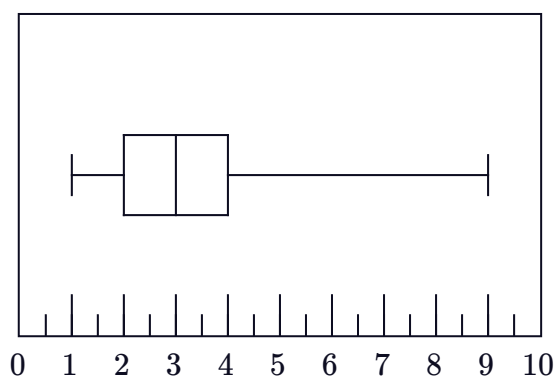
Box Plot 2



Histogram C



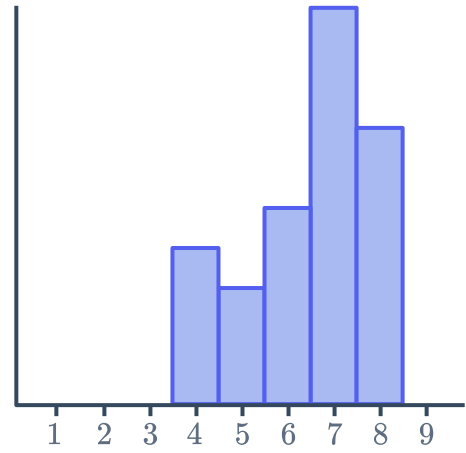
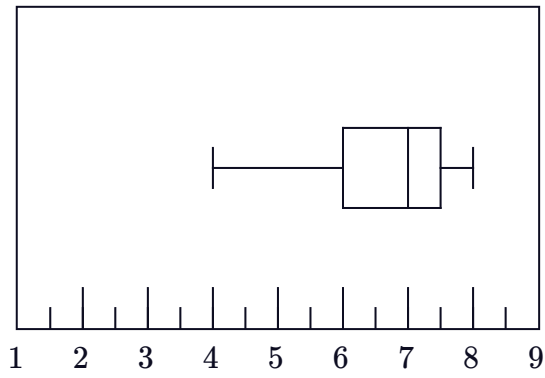
Box Plot 3





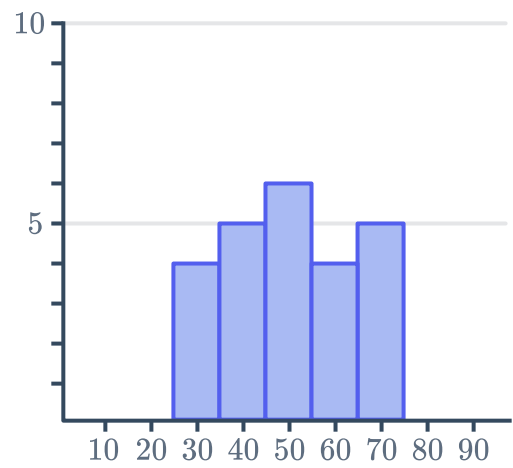
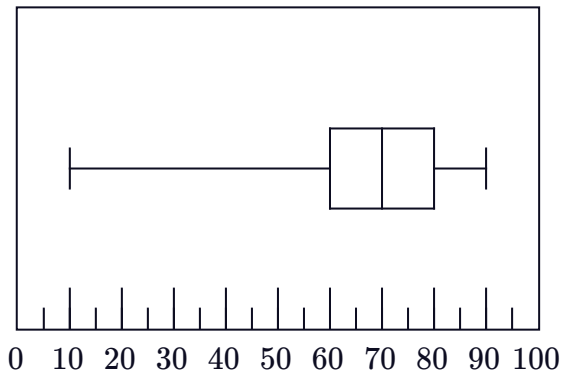
Histogram D

Box Plot 4



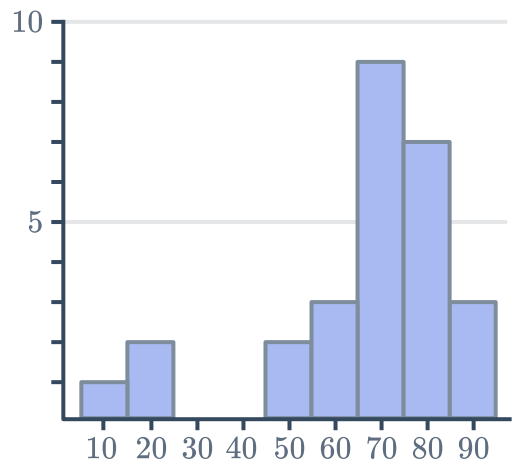
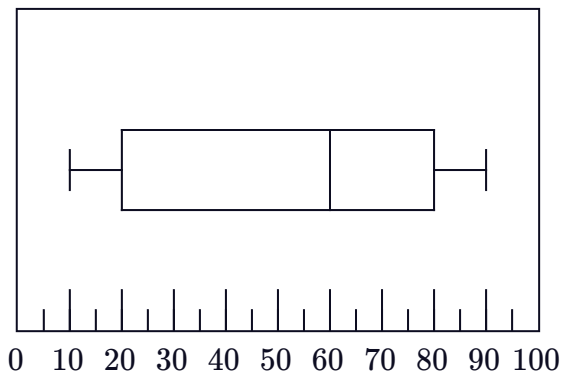
Histogram E

Box Plot 5



Histogram F

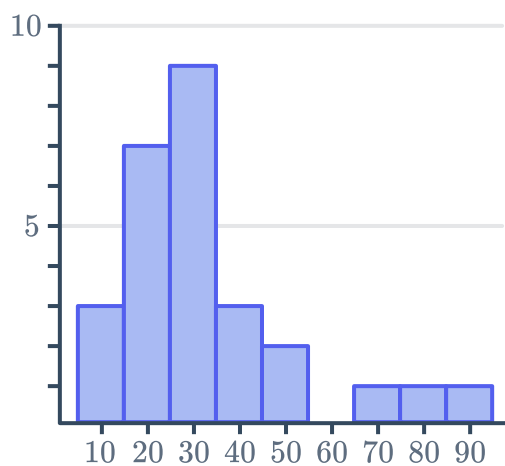
Box Plot 6



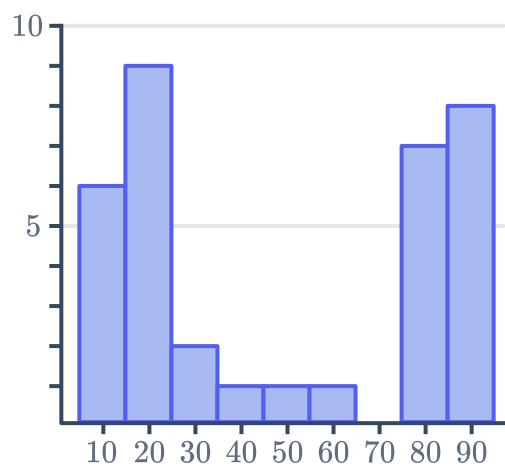
23 Construct a box plot for the following histograms:



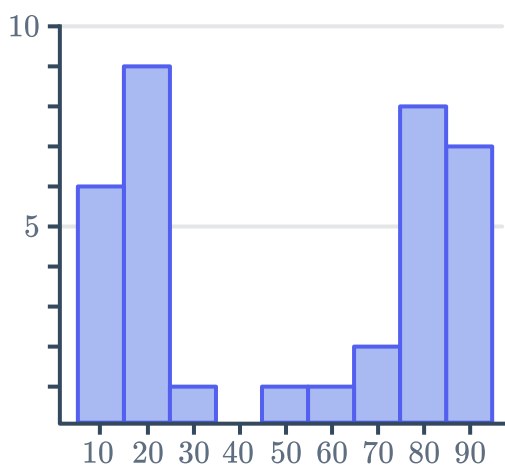
a



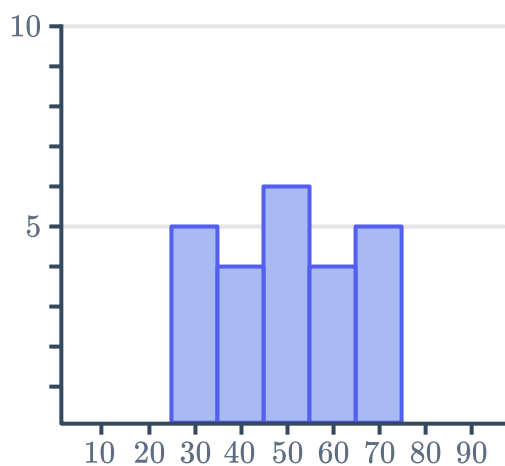
b



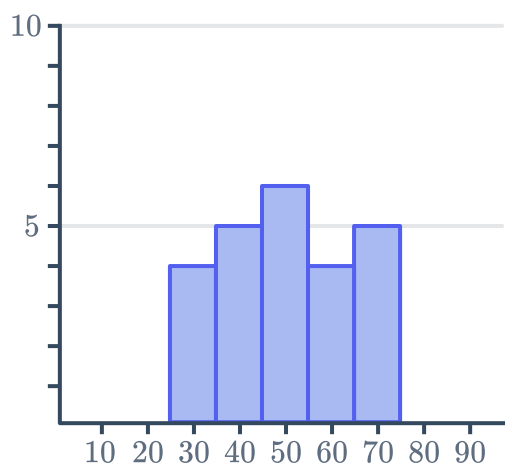
c



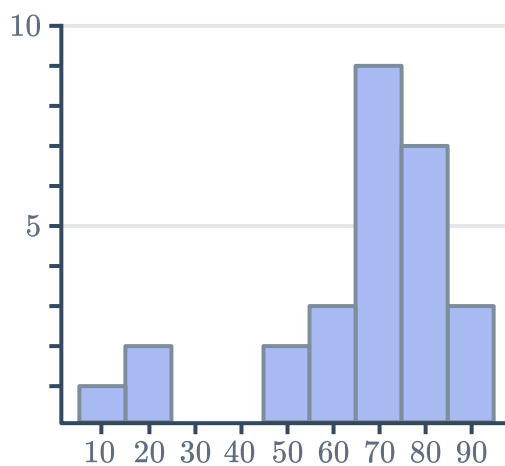
d



e



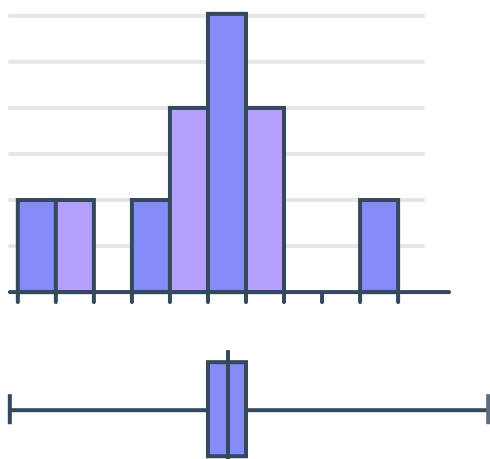
f



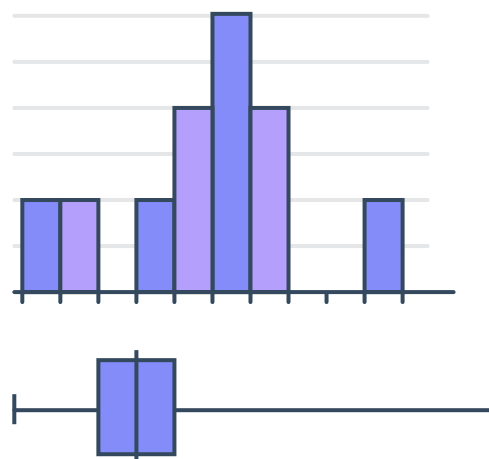
24 State whether the following pairs of histograms and box plots are correctly matched with respect to their shape:



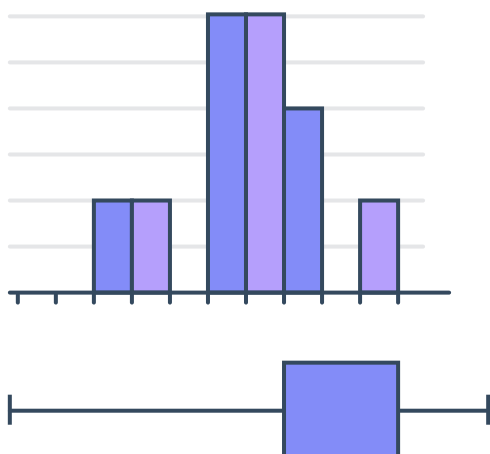
a



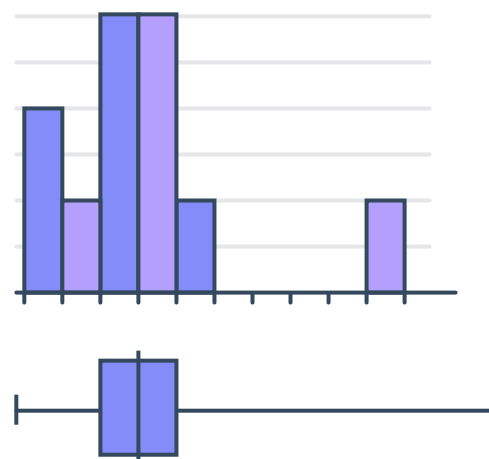
b



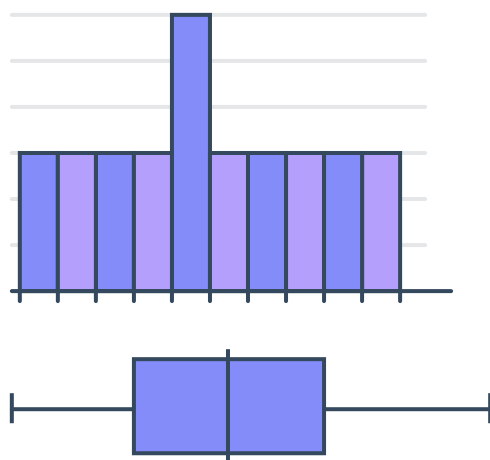
c



d



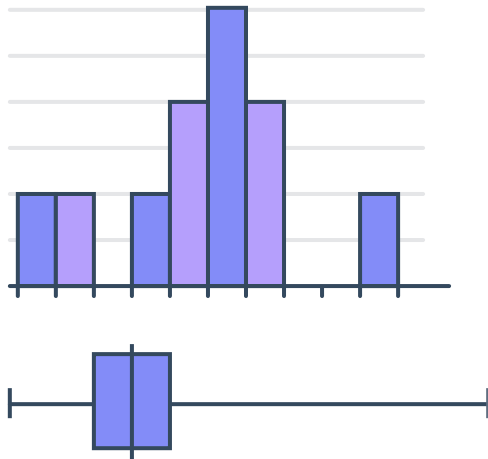
e



25 Explain why the following pairs of histograms and box plots do not match:



a



b

